

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
	$\vec{OA} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$, $\vec{OB} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, $\vec{OC} = a\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, $a < 0$ $\vec{AB} = \vec{BD}$, $ \vec{AB} = 4$		
(a)	E.g. $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB}$ or $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB} = \vec{OB} + \vec{OB} - \vec{OA} = 2\vec{OB} - \vec{OA}$ or $\vec{OD} = \vec{OB} + \vec{BD} = \vec{OB} + \vec{AB} = \vec{OA} + \vec{AB} + \vec{AB} = \vec{OA} + 2\vec{AB}$		
	$= \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ -5 \\ 7 \end{pmatrix}$ or $= \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + 2 \left(\begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -5 \\ 7 \end{pmatrix}$	M1	3.1a
	$= \begin{pmatrix} 6 \\ -7 \\ 10 \end{pmatrix} \text{ or } 6\mathbf{i} - 7\mathbf{j} + 10\mathbf{k}$	A1	1.1b
		(2)	
(b)	$(a-2)^2 + (5-3)^2 + (-2-4)^2$	M1	1.1b
	$\left\{ \vec{AC} = 4 \Rightarrow (a-2)^2 + (5-3)^2 + (-2-4)^2 = 4^2 \right.$ $\Rightarrow (a-2)^2 = 8 \Rightarrow a = \dots \text{ or } \Rightarrow a^2 - 4a - 4 = 0 \Rightarrow a = \dots$	dM1	2.1
	$(\text{as } a < 0 \Rightarrow) a = 2 - 2\sqrt{2} \text{ (or } a = 2 - \sqrt{8})$	A1	1.1b
		(3)	
(5 marks)			

Notes for Question	
(a)	
M1:	Complete <i>applied</i> strategy to find a vector expression for $\vec{OD}$
A1:	See scheme
Note:	Give M0 for subtracting the wrong way wrong to give e.g. $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) + (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}) - (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) = (4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) + (-2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}) = (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$
Note:	Writing e.g. $\vec{OD} = \vec{OB} + \vec{AB}$ or $\vec{OD} = 2\vec{OB} - \vec{OA}$ with no other work is M0
Note:	Finding <i>coordinates</i> , i.e. $(6, -7, 10)$ without reference to the correct position vectors is A0
Note:	Allow M1A1 for writing down $6\mathbf{i} - 7\mathbf{j} + 10\mathbf{k}$ with no working
Note:	M1 can be implied for at least two correct components in their position vector of D
(b)	
M1:	Finds the difference between $\vec{OA}$ and $\vec{OC}$, then squares and adds each of the 3 components Note: Ignore labelling
dM1:	Complete method of <i>correctly</i> applying Pythagoras' Theorem on $ \vec{AC} = 4$ and using a correct method of solving their resulting quadratic equation to find at least one of $a = \dots$
Note:	Condone at least one of either awrt 4.8 or awrt -0.83 for the dM mark
A1:	Obtains only one exact value, $a = 2 - 2\sqrt{2}$
Note:	Writing $a = 2 \pm 2\sqrt{2}$, without evidence of rejecting $a = 2 + 2\sqrt{2}$ is A0
Note:	Allow exact alternatives such as $2 - \sqrt{8}$ or $\frac{4 - \sqrt{32}}{2}$ for A1, and isw can be applied
Note:	Writing $a = -0.828\dots$, without reference to a correct exact value is A0

Q2.

Question Number	Scheme	Marks	AO's
	Attempts any one of $(\pm \overrightarrow{PQ} =) \pm (\mathbf{q} - \mathbf{p}), (\pm \overrightarrow{PR} =) \pm (\mathbf{r} - \mathbf{p}), (\pm \overrightarrow{QR} =) \pm (\mathbf{r} - \mathbf{q})$ Or e.g. $(\pm \overrightarrow{PQ} =) \pm (\overrightarrow{OQ} - \overrightarrow{OP}), (\pm \overrightarrow{PR} =) \pm (\overrightarrow{OR} - \overrightarrow{OP}), (\pm \overrightarrow{QR} =) \pm (\overrightarrow{OR} - \overrightarrow{OQ})$	M1	1.1b
	Attempts e.g. $\mathbf{r} - \mathbf{q} = 2(\mathbf{q} - \mathbf{p})$ $\mathbf{r} - \mathbf{p} = 3(\mathbf{q} - \mathbf{p})$ $\frac{2}{3}(\mathbf{q} - \mathbf{p}) = \frac{1}{3}(\mathbf{r} - \mathbf{q})$ $\mathbf{q} = \mathbf{p} + \frac{1}{3}(\mathbf{r} - \mathbf{p})$ $\mathbf{q} = \mathbf{r} + \frac{2}{3}(\mathbf{p} - \mathbf{r})$	dM1	3.1a
	E.g. $\Rightarrow \mathbf{r} - \mathbf{q} = 2\mathbf{q} - 2\mathbf{p} \Rightarrow 2\mathbf{p} + \mathbf{r} = 3\mathbf{q} \Rightarrow \mathbf{q} = \frac{1}{3}(\mathbf{r} + 2\mathbf{p})^*$	A1*	2.1
		(3)	
(3 marks)			

Notes:

M1: Attempts any of the relevant vectors by subtracting either way around. This may be implied by sight of any one of $\pm(\mathbf{q} - \mathbf{p}), \pm(\mathbf{r} - \mathbf{p}), \pm(\mathbf{r} - \mathbf{q})$ ignoring how they are labelled

dM1: Uses the given information and writes it correctly in vector form that if rearranged would give the printed answer

A1*: Fully correct work leading to the given answer. Allow $\mathbf{OQ} = \dots$ as long as $\mathbf{OQ}$ has been defined as $\mathbf{q}$ earlier.

In the working allow use of $\mathbf{P}$ instead of $\mathbf{p}$ and $\mathbf{Q}$ instead of $\mathbf{q}$ as long as the intention is clear.

Q3.

Question	Scheme	Marks	AOs
(a)	Attempts both $ \overrightarrow{PQ} = \sqrt{2^2 + 3^2 + (-4)^2}$ and $ \overrightarrow{QR} = \sqrt{5^2 + (-2)^2}$	M1	3.1a
	States that $ \overrightarrow{PQ} = \overrightarrow{QR} = \sqrt{29}$ so PQRS is a rhombus	A1	2.4
		(2)	
(b)	Attempts BOTH $\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = 7\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$ AND $\overrightarrow{QS} = -\overrightarrow{PQ} + \overrightarrow{PS} = 3\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$	M1	3.1a
	Correct $\overrightarrow{PR} = 7\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$ and $\overrightarrow{QS} = 3\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$	A1	1.1b
	Correct method for area PQRS. E.g. $\frac{1}{2} \times \overrightarrow{PR} \times \overrightarrow{QS} $	dM1	2.1
	$= \sqrt{517}$	A1	1.1b
		(4)	
(6 marks)			
Alt (b) Example using the cosine rule	Attempts $ \overrightarrow{QS} = \sqrt{3^2 + (-3)^2 + 2^2} = \sqrt{22}$ and so $22 = 29 + 29 - 2\sqrt{29}\sqrt{29} \cos SPQ$	M1	3.1a
	$\cos PQR = -\frac{18}{29}$ or $\cos SPQ = \frac{18}{29}$ Condone angles in degrees 51.6, 128.4 (1dp) or radians 2.24, 0.901 (3sf) here	A1	1.1b
	Correct method for area PQRS. E.g. $PQ \times QR \sin PQR = \sqrt{2^2 + 3^2 + (-4)^2} \times \sqrt{5^2 + (-2)^2} \times \frac{\sqrt{517}}{29}$	dM1	2.1
	$= \sqrt{517}$	A1	1.1b
		(4)	

FYI

$$\overrightarrow{QR} = 5\mathbf{i} + 0\mathbf{j} - 2\mathbf{k}$$

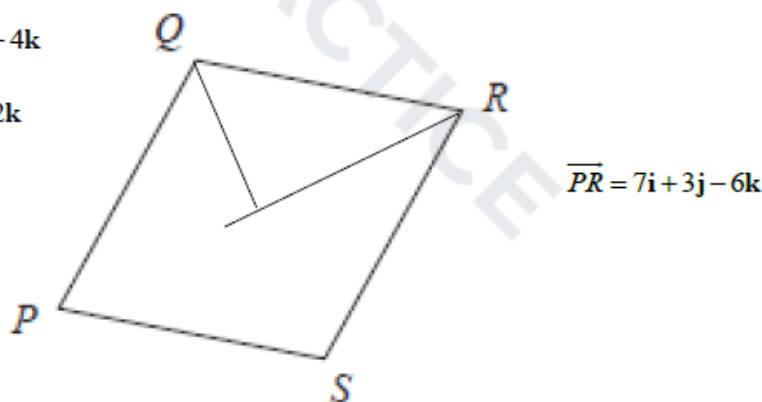
$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$$

$$\overrightarrow{SQ} = -3\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$$

$$\overrightarrow{MQ} = -1.5\mathbf{i} + 1.5\mathbf{j} - 1\mathbf{k}$$

M

$$\overrightarrow{PM} = 3.5\mathbf{i} + 1.5\mathbf{j} - 3\mathbf{k}$$



(a) Do not award marks in part (a) from work in part (b).

M1: Attempts both $|\overrightarrow{PQ}| = \sqrt{2^2 + 3^2 + (\pm 4)^2}$ and $|\overrightarrow{QR}| = \sqrt{5^2 + (\pm 2)^2}$ or PQ^2 and QR^2 . For this mark only, condone just the correct answers $|\overrightarrow{PQ}| = \sqrt{29}$ and $|\overrightarrow{QR}| = \sqrt{29}$. Alternatively attempts $\overrightarrow{PR} \bullet \overrightarrow{QS}$ or PM^2, MQ^2 and PQ^2 where M is the mid point of PR

A1: Shows that $|\overrightarrow{PQ}| = |\overrightarrow{QR}| = \sqrt{29}$ (with calculations) and states PQRS is a rhombus.

Condone poor notation such as $\overline{PQ} = \sqrt{29}$ here. So $\overline{PQ} = \overline{QR} = \sqrt{29}$ hence rhombus.

Requires both a reason and a conclusion. The reason may be given at the start of their solution.

In the alternatives $\overline{PR} \bullet \overline{QS} = (7i + 3j - 6k) \bullet (3i - 3j + 2k) = 21 - 9 - 12 = 0$ so diagonals cross at

90° so PQRS is a rhombus or $PM^2 + MQ^2 = PQ^2 = 23.5 + 5.5 = 29 \Rightarrow \angle PMQ = 90^\circ \Rightarrow$ Rhombus

(b) Candidates can transfer answers from (a) to use in part (b) to find the area

Look through their complete solution first. The first two marks are for finding the elements that are required to calculate the area. The second set of two marks is for combining these elements correctly. If the method is NOT shown on how to find vector it can be implied by two correct components. Allow as column vectors.

M1: For a key step in solving the problem. It is scored for attempting to find both key vectors.

Attempts both $\overline{PR} = \overline{PQ} + \overline{QR} = (7i + 3j - 6k)$ AND $\overline{QS} = -\overline{PQ} + \overline{PS} = (3i - 3j + 2k)$

You may see $\overline{PM} = \frac{1}{2}\overline{PQ} + \frac{1}{2}\overline{QR} = \left(\frac{7}{2}i + \frac{3}{2}j - 3k\right)$ AND $\overline{QM} = -\frac{1}{2}\overline{PQ} + \frac{1}{2}\overline{PS} = \left(\frac{3}{2}i - \frac{3}{2}j + 1k\right)$

A1: Accurately finds both key vectors whose lengths are required to solve the problem.

Score for both $\overline{PR} = 7i + 3j - 6k$ and $\overline{QS} = 3i - 3j + 2k$ (Allow either way around.)

or both $\overline{PM} = \frac{7}{2}i + \frac{3}{2}j - 3k$ and $\overline{QM} = \frac{3}{2}i - \frac{3}{2}j + 1k$ (Allow either way around.)

dM1: Constructs a rigorous method leading to the area PQRS. Dependent upon previous M.

E.g. See scheme. Alt: the sum of the area of four right angled triangles e.g. $4 \times \frac{1}{2} \times |\overline{PM}| \times |\overline{QM}|$.

A1: $\sqrt{517}$

Alternatives for (b). Two such ways are set out below

Alt 1-Examples via cosine rule but you may see use of scalar product via a Further Maths method.

M1: For a key step in solving the problem. In this case it for an attempt at $\cos PQR$ or $\cos SPQ$.

Don't be too concerned with the labelling of the angle which may appear as θ .

$$\text{Attempts } \pm \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} \bullet \pm \begin{pmatrix} 5 \\ 0 \\ -2 \end{pmatrix} = \sqrt{2^2 + 3^2 + (-4)^2} \times \sqrt{5^2 + (-2)^2} \cos PQR$$

A1: Finds the cosine of one of the angles in the Figure.

Look for $\cos \dots = -\frac{18}{29}$ or $\cos \dots = \frac{18}{29}$ which may have been achieved via the cosine rule.

Accept rounded answers and the angles in degrees 51.6, 128.4 (1dp) or radians 2.24, 0.901 (3sf) here.

dM1: Constructs a rigorous method leading to the area PQRS. Implied by awrt 22.7

$$\text{E.g. } PQ \times QR \sin PQR = \sqrt{2^2 + 3^2 + (-4)^2} \times \sqrt{5^2 + (-2)^2} \times \frac{\sqrt{517}}{29}$$

A1: $\sqrt{517}$

Alt 2-Example via vector product via a Further Maths method.

M1: For a key step in solving the problem. In this case it for an attempt at $\pm \overline{PQ} \times \overline{QR}$

$$\text{E.g. } \overline{PQ} \times \overline{QR} = \begin{vmatrix} i & j & k \\ 2 & 3 & -4 \\ 5 & 0 & -2 \end{vmatrix} = (3 \times -2 - 0 \times -4)i - (2 \times -2 - 5 \times -4)j + (2 \times 0 - 3 \times 5)k$$

A1: E.g. $\overline{PQ} \times \overline{QR} = -6i - 16j - 15k$

dM1: Constructs a rigorous method leading to the area PQRS. In this case $|\overline{PQ} \times \overline{QR}|$

A1: $= \sqrt{(-6)^2 + (-16)^2 + (-15)^2} = \sqrt{517}$

Q4.

Question	Scheme	Marks	AOs
(a)	$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = -3\mathbf{i} - 4\mathbf{j} - 5\mathbf{k} + \mathbf{i} + \mathbf{j} + 4\mathbf{k} = \dots$	M1	1.1b
	$= -2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$	A1	1.1b
		(2)	
(b)	At least 2 of $(AC^2) = 2^2 + 3^2 + 1^2, (AB^2) = 3^2 + 4^2 + 5^2, (BC^2) = 1^2 + 1^2 + 4^2$	M1	1.1b
	$2^2 + 3^2 + 1^2 = 3^2 + 4^2 + 5^2 + 1^2 + 1^2 + 4^2 - 2\sqrt{3^2 + 4^2 + 5^2}\sqrt{1^2 + 1^2 + 4^2} \cos ABC$	M1	3.1a
	$14 = 50 + 18 - 2\sqrt{50}\sqrt{18} \cos ABC$ $\Rightarrow \cos ABC = \frac{50 + 18 - 14}{2\sqrt{50}\sqrt{18}} = \frac{9}{10}^*$	A1*	2.1
		(3)	
(b) Alternative			
	$AB^2 = 3^2 + 4^2 + 5^2, BC^2 = 1^2 + 1^2 + 4^2$	M1	1.1b
	$\overrightarrow{BA} \cdot \overrightarrow{BC} = (3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}) \cdot (\mathbf{i} + \mathbf{j} + 4\mathbf{k}) = 27 = \sqrt{3^2 + 4^2 + 5^2}\sqrt{1^2 + 1^2 + 4^2} \cos ABC$	M1	3.1a
	$27 = \sqrt{50}\sqrt{18} \cos ABC \Rightarrow \cos ABC = \frac{27}{\sqrt{50}\sqrt{18}} = \frac{9}{10}^*$	A1*	2.1
(5 marks)			
Notes			

(a)

M1: Attempts $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

There must be attempt to add not subtract.

If no method shown it may be implied by two correct components

A1: Correct vector. Allow $-2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ and $\begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}$ but not $\begin{pmatrix} -2\mathbf{i} \\ -3\mathbf{j} \\ -1\mathbf{k} \end{pmatrix}$

(b)

M1: Attempts to "square and add" for at least 2 of the 3 sides. Follow through on their $\overrightarrow{AC}$

Look for an attempt at either $a^2 + b^2 + c^2$ or $\sqrt{a^2 + b^2 + c^2}$

M1: A correct attempt to apply a correct cosine rule to the given problem; Condone slips on the lengths of the sides but the sides must be in the correct position to find angle ABC

A1*: Correct completion with sufficient intermediate work to establish the printed result.

Condone different labelling, e.g. $ABC \leftrightarrow \theta$ as long as it is clear what is meant

It is OK to move from a correct cosine rule $14 = 50 + 18 - 2\sqrt{50}\sqrt{18} \cos ABC$

via $\cos ABC = \frac{54}{2\sqrt{50}\sqrt{18}}$ o.e. such as $\cos ABC = \frac{(5\sqrt{2})^2 + (3\sqrt{2})^2 - (\sqrt{14})^2}{2 \times 5\sqrt{2} \times 3\sqrt{2}}$ to $\cos ABC = \frac{9}{10}$

Alternative:

M1: Correct application of Pythagoras for sides AB and BC or their squares

M1: Recognises the requirement for and applies the scalar product

A1*: Correct completion with sufficient intermediate work to establish the printed result